ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CIT5611 CYBER LAW AND ETHICS

END OF SEMESTER EXAMINATION

MAY/JUNE 2026

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

A public university in Zambia has adopted several digital systems, including an online learning platform, cloud-based student records system, biometric access control, institutional email, and online payment services. Recently, the university experienced a cyber incident. Attackers gained unauthorised access to the student records system and obtained student numbers, email addresses, grades, internal messages, and payment records.

At the same time, some students have been sharing unverified information about the breach on social media. Some posts accuse named university officials of misconduct without evidence. Other users have reposted insulting and abusive comments. Management wants to delay public communication until investigations are complete, arguing that immediate disclosure may damage the university's reputation.

The university also discovers that one department has been using unlicensed software, some staff members have uploaded confidential institutional documents into public AI tools, and a vendor hosting part of the system stores data outside Zambia.

Using the above scenario, answer the following questions:

a) Apply any three ethical theories to evaluate the conduct of the university, staff, students, and attackers in the scenario. **[10 Marks]**
b) Discuss the role of government regulation in addressing incidents of this nature. In your answer, refer to relevant areas of law. **[10 Marks]**
c) Examine the legal and ethical duties of the university after the breach. Your answer should address breach notification, protection of affected students, evidence preservation, vendor risk management, and public communication. **[10 Marks]**
d) Assess how freedom of expression should be balanced with content controls in relation to the students' social media posts. In your answer, distinguish between lawful criticism, misinformation, online harassment, reputational harm, and content that may require lawful restriction. **[10 Marks]**
e) Evaluate the legal and ethical risks arising from the use of biometrics, cloud computing, AI tools, and online payment platforms by the university. Propose practical governance controls that should be implemented. **[10 Marks]**

(50 Marks)

SECTION B: Attempt any TWO (2) questions in this section

Question 2

The Internet is global, fast-moving, decentralised and difficult for one State to regulate alone. At the same time, governments are expected to protect citizens, critical infrastructure, public order, personal data, and national security.

a) Explain why government regulation of the Internet has increased over time. **[5 marks]**
b) Discuss the main areas of law that support Internet regulation in Zambia. **[7 marks]**
c) Examine the constitutional tension between freedom of expression, privacy, public order, national security, and protection of others in cyberspace. **[6 Marks]**

d) Discuss the importance of inter-government collaboration in cyber regulation, especially in relation to cybercrime investigations, digital evidence, mutual legal assistance, harmonisation of laws, and protection of critical information infrastructure. [7 **marks]**

(25 marks)

Question 3

Digital technologies make it easy to create, copy, share, remix, modify, and distribute content across borders. This creates opportunities for innovation but also raises legal and ethical problems.

a) Explain the meaning of intellectual property in cyberspace and distinguish between copyright, trademarks, patents, and trade secrets. **[6 marks]**
b) Discuss the main challenges of protecting intellectual property online, including piracy, plagiarism, counterfeiting, unauthorised downloads, software misuse, and cross-border enforcement difficulties. **[6 marks]**
c) Examine the legal and ethical issues involved in software ownership, open-source software, software licensing, and use of unlicensed software in public institutions. **[6 Marks]**
d) Discuss domain names, cybersquatting, and typosquatting as intellectual property issues in cyberspace. **[4 marks]**
e) Assess how cyber law should balance creator reward, public access to knowledge, innovation, digital freedom, and fair use of digital content. **[3 marks]**

(25 marks)

Question 4

Artificial intelligence and other digital tools are increasingly influencing education, employment, finance, policing, admissions, and public service delivery. In education, students can now use AI tools to generate essays, solve problems, and support learning, forcing institutions to rethink what counts as original work and academic honesty. In other sectors, AI systems are used to make or support decisions that can affect people's access to jobs, loans, education, and public services. When such systems produce biased, unfair, or harmful outcomes, difficult ethical questions arise about accountability, fairness, transparency, and human oversight.

a) Should schools, colleges, and universities in Zambia treat undisclosed use of AI tools in assignments as academic dishonesty, or should they redesign assessment methods to reflect the AI era? Discuss. **(8 marks)**
b) If an AI system used in hiring, school admissions, policing, lending, or public services produces biased outcomes, who should be held ethically accountable: the developer, the institution using it, or the regulator? **(9 marks)**
c) Suggest ethical principles and practical safeguards that should guide the use of AI in education and public decision-making. **(8 marks)**

(25 marks)

END OF EXAMINATION